BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Primary Storage: RAM, ROM and Cache

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

SILENT

BLACK PEN: Answer all four before looking at the model answers.

1. Previous lesson: what is lossless compression?

2. How much is saved if 2000 KB becomes 1500 KB?

3. Convert 32 bits into bytes.

4. Why must executable code be restored without lost data?

Today's objectives

Distinguish the roles of RAM, ROM and cache, including volatility.

Key vocabulary - Read, cover, recall

RAM
Volatile store for running programs

ROM
Non-volatile store for firmware

cache
Small, fast storage near the CPU

volatile
Loses contents without power

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Three kinds of primary storage

Primary storage supports the CPU and the programs it runs.

RAM	ROM	CACHE
Running apps and data	Start-up instructions	Frequently used CPU data
Volatile	**Non-volatile**	**Small and very fast**

Volatile means stored contents are lost when electrical power stops.

I Do - Choose the right memory

LISTEN

Opening a game: instructions/data are loaded into RAM.

Starting the computer: firmware is retained in ROM.

Repeated CPU instructions: small fast cache helps access.

We Do - Guided practice

MWBs

Match each task: a running app, start-up firmware and frequently used CPU data.

A running app belongs in: _____

Start-up firmware belongs in: _____

Repeated CPU data belongs in: _____

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

MWBs

Start on paper. Do not look at model answers until page 4.

```
capacity = 8    used = 3
print(capacity - used)
```

Predict the number of gigabytes available.

Change used to 5. What does the program display?

Improve the code to show how much remains when used is 7.

You Do - Independent work

SILENT

Explain why RAM is called volatile. [2]

Name the primary store commonly used for firmware. [1]

Give one way cache differs from RAM. [2]

Thinking: Why can more RAM help someone running many apps?

Exit Ticket - True or False?

THUMBS

Answer before you turn to the green-pen page.

1. RAM normally loses its contents when power is switched off. ☐ T ☐ F
2. ROM usually stores firmware and retains its contents. ☐ T ☐ F
3. CPU cache is generally larger and slower than RAM. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick correct answers. For a mistake, use the model answer then copy the FULL question and write a fresh answer below.

Do Now - Answers

Q1 The original data can be restored exactly. ☐

Q2 500 KB. ☐

Q3 32 / 8 = 4 bytes. ☐

Q4 Missing or changed bits may break the program. ☐

We Do - Guided practice

App: RAM; firmware: ROM; often-used CPU data: cache

Key vocabulary

RAM ☐
Volatile store for running programs

ROM ☐
Non-volatile store for firmware

cache ☐
Small, fast storage near the CPU

volatile ☐
Loses contents without power

Code First - Answers

GREEN PEN

Predict ☐

8 - 3 = 5 GB

Write ☐

8 - 5 = 3 GB

MWBs

Improve ☐

Set used = 7: output is 1 GB

Exit Ticket - Explanations

1. True

2. True

3. False: cache is smaller and faster

You Do - Self-mark

GREEN PEN

Q1	Its stored contents are lost when power is removed	☐
Q2	ROM	☐
Q3	Cache is smaller and faster; it holds frequently needed data	☐
Thinking	More working data can remain in RAM instead of slower storage.	☐

Try the question again

GREEN PEN

1. Copy the FULL question you got wrong (or the hardest):

2. In GREEN PEN, write a NEW answer. Show working: